



How We Tested

Our test-bed comprised an Intel Pentium 4 3 GHz processor, running on an Intel D875PBZ motherboard, with 512 MB of DDR 333 RAM, a Seagate Barracuda 120 GB SATA hard drive and an Ennyah GeForce FX 5700 graphics card.

All the monitors were tested for performance and features. The features and performance parameters are allotted 40 per cent and 60 per cent respectively. The performance and features scores were added, and the sum was divided by cost. This gave us an index for value for money.

Performance: We used the DisplayMate benchmarking software to gauge performance. It uses images to test for geometrical accuracy. This aspect is especially important for CAD and CAM designers, and users of graphics applications.

Another important factor to consider is the ability of the monitor to hold image shape in a stable manner. Sudden changes in brightness levels can cause distortion, say, when you toggle between an application and the desktop.

In the sharpness and resolution set of tests, we tested the image sharpness offered by the monitor and the resolution at which it displays pictures clearly. The Defocusing, Blooming and Halos tests showed us how much thicker an image looks compared to its darker counterpart: in this test, the image detail projected by the CRT is measured.

The pixel persistence test was run specifically for the LCDs, to check for jerks or image blurring. Another LCD-specific test was the viewable angle test. Unlike CRT monitors, when you view an LCD screen from an angle, things are not as bright and clear. We tested to find the angles at which the screen still looked crisp.

Features: We looked for value-additions and other aspects that aimed at increasing the user-friendliness of the display. The features noted varied in their significance—from simple things such as ease of use of the OSD, to power consumption for LCDs, and depth and weight for CRTs.

We also looked for, and awarded points to, important parameters such as pixel size and video bandwidth, in both LCD and CRT monitors. A smaller pixel size and higher video bandwidth was given preference, as it helps to make the display crisper and the colours more vivid.

Even the monitor bases were tested, to see how much they could be tilted—the more the better.

Wall-mounted LCDs are convenient, and earned more points. Height adjustment was also considered. Interestingly, some could be rotated from landscape to portrait mode.

User manuals are often overlooked, as is the quick start guide. We pored through them to see how helpful they'd be if we didn't already know how to set up a monitor.



CATEGORY

CRT MONITORS 17"

Brand	BenQ	BenQ	HCL	HCL	LG	LG	LG	Microtek
Model	G775	V773	HCM780M	HCM785RFM	Flatron E700BH	Flatron E700s	Studioworks 700S	Flatvision 43F1
Size (Inches)	17	17	17	17	17	17	17	17
Viewable Area (Inches)	16	16	16	16	16	16	16	16
F&ST (Y/N)	Y	N	N	Y	Y	Y	N	N
Dot Pitch (mm) (Diagonal)	0.25	0.27	0.27	0.21 (horizontal)	0.25	0.25	0.23 (horizontal)	0.28
Vertical Scan Rate (Hz)	50 - 120	50 - 120	50 - 160	50 - 160	50 - 160	51 - 160	50 - 160	50 - 120
Horizontal Scan Rate (KHz)	30 - 72	30 - 72	30 - 70	30 - 72	30 - 71	31 - 71	30 - 70	30 - 70
Maximum Resolution/Refresh Rate (Hz)	1280 x 1024 @ 60	1280 x 1024 @ 67	1280 x 1024 @ 60	1280 x 1024 @ 60	1280 x 1024 @ 60	1280 x 1024 @ 60	1280 x 1024 @ 60	1280 x 1024 @ 60
Video Bandwidth (MHz)	110	110	110	110	110	110	NA	110
CRT type	ShadowMask	ShadowMask	ShadowMask	ShadowMask	ShadowMask	ShadowMask	ShadowMask	ShadowMask
W x H x D (mm)	406 x 422 x 417.5	406 x 422 x 412.5	410 x 402 x 420	410 x 402 x 420	400 x 395 x 424	401 x 395 x 424	400 x 420 x 395	406 x 404 x 423
Weight (Kg)	16.4	15.4	15.5	16	16	16	14.4	15.5
In-built Speakers (Y/N)	N	N	N	N	N	N	N	N
Anti-Glare Coating (Y/N)	Y	N	Y	Y	Y	Y	Y	Y
Bundled Accessories (Manuals, Colour Profile, Driver and Calibrating software, etc.)	User manual, driver CD	User manual, driver CD	User manual	User manual	User manual, driver CD	User manual, driver CD	User manual, Driver CD	User manual
Performance								
OSD Usability (scale of 5)	3.25	3.5	3.75	4	3.5	3.5	3.5	4.5
DisplayMate Tests								
Geometry test (scale of 25)	6.5	8.5	5.25	4.25	6.5	7.75	8.75	4
Sharpness and Resolution (scale of 35)	23	22.5	17	22.5	24.5	23.5	22.25	20.75
Screen Pixel Resolution (scale of 5)	3.5	3.75	2.75	3.5	3.75	3.75	3.75	3.25
Colour and Grey Scale (scale of 45)	34	30	29	28.5	28	31.5	30.25	30.5
Miscellaneous Effects (scale of 25)	15.25	15.5	16.75	16	15	16.25	16.75	16
Features	17	10	14	19	21	21	15	11
Performance	34.25	33.85	29.5	31	32.4	34.7	34.45	31.65
Overall (Features + Performance)	51.25	43.85	43.5	50	53.4	55.7	49.45	42.65
Value for Money	2.26	2.53	2.67	2.39	2.25	2.49	2.81	2.81
Price	8,500	6,500	6,100	7,850	8,900	8,400	6,600	5,700
Grade	B+	B+	B+	B+	B+	B+	B+	B+